

■ CodeGraph.AI — Developer Setup Guide

CodeGraph.AI is an AI-powered developer assistant that integrates **VS Code**, **Jira**, and **GitHub**.

It intelligently analyzes your codebase, reads Jira stories, and automates pull request generation through a FastAPI backend.

This guide explains how to set up and use CodeGraph.AI on a new laptop.

■■ 1■■ Prerequisites

Before starting, ensure you have these installed:

- ■ Python 3.9+
- ■ pip or conda
- ■ Visual Studio Code
- ■ Node.js (for npm and VSCE)
- ■ Internet access (for Jira and GitHub integrations)

■ 2■■ Backend (FastAPI) Setup

Step 1 — Install dependencies

Open a terminal in the backend project folder and run:

```
pip install -r requirements.txt
```

If requirements.txt is not available, manually install:

```
pip install fastapi uvicorn requests pydantic
```

Step 2 — Start the FastAPI Server

From the root of the backend (where main.py is located), run:

```
uvicorn main:app --reload --host 127.0.0.1 --port 8000
```

■ Expected output:

```
INFO: Uvicorn running on http://127.0.0.1:8000
```

```
INFO: Application startup complete.
```

Verify it's running by visiting: <http://127.0.0.1:8000>

You should see:

```
{"status": "CodeGraph.AI backend is running"}
```

■ 3 ■ VS Code Extension Installation

The CodeGraph.AI VS Code extension (.vsix file) allows developers to run the agent directly inside VS Code.

The .vsix file is included in this repository (e.g. codegraph-ai-0.0.1.vsix).

Option A — Install via Command Line

```
code --install-extension ./codegraph-ai-0.0.1.vsix
```

Example:

```
code --install-extension C:\Projects\CodeGraph_AI\codegraph-ai-0.0.1.vsix
```

Option B — Install via VS Code Interface

1. Open Visual Studio Code
2. Go to Extensions (Ctrl + Shift + X)
3. Click the "⋮" (three dots) menu in the top-right corner
4. Choose "Install from VSIX..."
5. Select the codegraph-ai-0.0.1.vsix file
6. Wait for the confirmation that installation was successful

■ 4■■ Running the Extension

1. Open your main repository (where you want CodeGraph.AI to make changes)
2. Ensure FastAPI backend is running (<http://127.0.0.1:8000>)
3. In VS Code, press Ctrl + Shift + P
4. Type and select: Run CodeGraph.AI Agent
5. Enter Jira ID and Branch name
6. The extension will call the FastAPI endpoint /run-agent and return results

■■ 5■■ Configuration (Optional)

You can configure the backend URL in VS Code settings:

Settings → Extensions → CodeGraph.AI → Backend URL

Example: <http://127.0.0.1:8000> or <https://api.codegraph.ai>

■ 6■■ Updating the Extension

```
code --uninstall-extension codegraph-ai
```

```
code --install-extension ./codegraph-ai-0.0.2.vsix
```

■ 7■■ Verifying Setup

Check Backend: `http://127.0.0.1:8000`

Check Extension: ensure CodeGraph.AI is in VS Code extensions

Test Command: Run CodeGraph.AI Agent and check FastAPI terminal logs

■ 8■■ Common Commands

Start FastAPI: `uvicorn main:app --reload --host 127.0.0.1 --port 8000`

Install Extension: `code --install-extension ./codegraph-ai-0.0.1.vsix`

Uninstall Extension: `code --uninstall-extension codegraph-ai`

List Extensions: `code --list-extensions`

■ 9■■ Team Setup Workflow

1. Push repo (including .vsix file)
2. Clone repo
3. Install Python dependencies
4. Start FastAPI backend
5. Install .vsix extension
6. Run CodeGraph.AI Agent

■■■ Author

****Shanaka Chathuranga****

Creator of CodeGraph.AI

Empowering developers with intelligent, agent-based automation for Jira and GitHub workflows.